

Skolem IRI Pattern for the XSD → RDF Architecture

1. Purpose of This Document

This document defines the Skolem IRI pattern used in the XSD → RDF transformation architecture. It explains:

- What a Skolem IRI is
- Why Skolem IRIs are required in this project
- When they must be used
- The deterministic minting pattern adopted
- How they support the Reversible Core / Derived Views principle

This specification is architectural and normative for the core transformation layer.

2. What Is a Skolem IRI?

In RDF, a blank node represents an unnamed resource — something that exists structurally but does not have a global identifier.

A **Skolem IRI** is a system-minted, deterministic IRI that replaces a blank node.

Instead of this:

```
[ a xsd2rdf:AttributeGroupUse ;  
  xsd2rdf:refQName "AttributeExtension" ]
```

We mint a stable IRI such as:

```
<https://example.org/id/.../_skolem/attributeGroupUse/1>
```

A Skolem IRI:

- Is generated by the transformation system
 - Is deterministic
 - Is not globally meaningful outside the system
 - Exists solely to provide stable identity to otherwise anonymous structural nodes
-

3. Why Blank Nodes Are Insufficient for This Project

This XSD → RDF architecture requires:

- Deterministic graph generation
- Stable test results
- Diff-friendly serialization
- Full structural reversibility back to XSD
- Precise traceability of structural occurrences

Blank nodes break determinism because:

- Their identifiers are not stable across serializations
- They cannot be referenced predictably
- They complicate test assertions
- They prevent exact structural comparison

Therefore, the Reversible Core must not rely on blank nodes for structural components.

4. When Skolem IRIs Must Be Used

Skolem IRIs are required for **structural occurrence nodes** — nodes that:

- Exist in the schema structure
- Do not have a global `name=` attribute
- Must be preserved for round-trip fidelity

Examples include:

- `xs:attributeGroup` use sites (`ref=` occurrences)
- `xs:group` use sites
- Model group containers (`sequence` , `choice` , `all`)
- Particle nodes
- Anonymous complex types
- Identity constraint applications
- Any structural node without global identity

Skolem IRIs are **not used** for:

- Globally named components (`xs:complexType name=...`)
 - Globally named elements
 - Globally named attributeGroups
 - Any component with stable QName identity
-

5. Named Component IRI Minting (Non-Skolem)

Named global components are minted using a deterministic rule based on:

- The ontology base IRI
- The XSD component kind
- The targetNamespace
- The NCName

Pattern:

```
{base}/xsd/{componentKind}/{urlenc(targetNamespace)}#{NCName}
```

Example:

```
{base}/xsd/attributeGroup/{urlenc(http://www.mismo.org/residential/2009/schemas)}#AttributeExtension
```

These IRIs represent definitions and are stable across all builds.

6. Skolem IRI Minting Pattern

For structural occurrence nodes, the following Skolem pattern is adopted:

```
{parentIRI}/_skolem/{kind}/{path}/{ordinal}
```

Where:

- `{parentIRI}` = IRI of the containing component
- `{kind}` = structural node type (e.g., `attributeGroupUse`)
- `{path}` = stable structural token path
- `{ordinal}` = 1-based position among same-kind siblings

6.1 Path Token Convention

The `{path}` segment is a tokenized structural path, not raw XPath.

Recommended tokens:

- `ct` - complexType root
- `seq` - sequence
- `choice` - choice

- `all` – all compositor
- `attrs` – attribute section
- `attrGroup` – attributeGroup use context

Example path:

```
ct/attrs/attrGroup
```

6.2 Example: MISMO Container Pattern

Given a complexType `Loan` with two attributeGroup references:

1. `ref="xlink:MISMOresourceLink"`
2. `ref="AttributeExtension"`

If the parent complexType IRI is:

```
{base}/xsd/complexType/{urlenc(mismoNS)}#Loan
```

The Skolem IRIs are:

```
{parentIRI}/_skolem/attributeGroupUse/ct/attrs/attrGroup/1
{parentIRI}/_skolem/attributeGroupUse/ct/attrs/attrGroup/2
```

These IRIs:

- Preserve ordering
- Scope identity to the parent
- Provide stable graph anchors

7. Required Properties on Skolem Use Nodes

Each Skolem occurrence node must record:

- `refQNameLexical` – original lexical QName string
- `refNamespaceURI` – resolved namespace URI
- `refLocalName` – local name
- `ordinal` – positional index
- `resolvesTo` – IRI of definition (if known)

This guarantees that even unresolved references can be round-tripped.

8. Determinism Requirements

The Skolem generation algorithm must:

1. Be purely mechanical
2. Depend only on structural position
3. Produce identical IRIs for identical input
4. Not depend on runtime memory state
5. Not depend on processing order across files

If two builds of the same schema produce different Skolem IRIs, the implementation is incorrect.

9. Relationship to the Reversible Core Principle

The Reversible Core requires that:

- All structural information necessary to reconstruct the XSD is preserved
- No structural information is flattened or collapsed
- Occurrence ordering is retained

Skolem IRIs are essential because they:

- Give identity to structural nodes
- Preserve ordering
- Enable explicit graph linking
- Allow precise test assertions
- Support derived semantic views without altering the core

Derived views may collapse or flatten structure, but the Core must remain fully structural.

10. Architectural Summary

Skolem IRIs provide:

- Deterministic structural identity
- Test stability
- Round-trip safety
- Diff-friendly RDF output
- Clean separation between definition identity and structural occurrence identity

They are mandatory for structural nodes lacking global names and are a foundational element of the XSD → RDF mechanical transformation layer.

This pattern is normative for all structural component extraction going forward.